

肝臟病理影像 AI 模型效能驗證與臨床應用評估

Performance Validation and Clinical Application Assessment of AI Models in Liver Pathology Imaging

一、摘要

隨著人工智慧技術迅速發展，AI 在醫療影像分析領域應用愈加廣泛，特別是在輔助醫療診斷方面成果顯著，此技術對於提升醫學影像解讀的精確性具有重要作用，有助於診斷的準確性與治療模式的優化。本研究聚焦於肝臟疾病，尤其針對非酒精性脂肪肝病 (NAFLD)、原發性肝癌 (HCC)、肝纖維化及肝臟免疫反應，從單純脂肪堆積到進一步的肝纖維化、肝硬化，乃至於肝癌的病程發展。本研究計畫將應用 AI 及數位病理技術深入分析肝臟細胞層級的病理特徵，提供精確的量化數據，協助病理醫師進行臨床診斷，以期提高患者的治療效果及生活品質。

二、簡介

本計畫採用臺北榮民總醫院的回溯性資料進行 AI 模型的驗證與分析，內容涵蓋肝臟組織病理影像、病歷紀錄及報告。計畫所使用的病理 AI 模型，是以成大醫院的檢體資料進行訓練開發，再透過臺北榮民總醫院的檢體資料進行獨立驗證，其目的是藉由分析肝臟病理影像中由病理 AI 模型量化的數據及所顯示的判讀結果，輔助病理醫師診斷，減少病理醫師之間的主觀差異，進而提升診斷準確性與臨床應用價值。本研究將聚焦於脂肪浸潤、纖維化、免疫反應、肝炎及肝癌等病理特徵，並進行以下三項驗證：

- 模型偵測準確度驗證
- 模型數據與病理報告的相關性分析
- 模型數據對醫師診斷一致性與準確度提升的效果驗證

各 AI 模型將依其發展階段，依序進行部分或全部的驗證試驗，以確保其在不同醫院資料下的適用性與可靠性。

三、試驗目的

本研究意旨在驗證 AI 技術於肝臟病理影像判讀之應用，特別針對 NAFLD (非酒精性脂肪肝)、肝纖維化、肝炎及 HCC (肝細胞癌) 等疾病進行病理分析。先前已使用成大醫院病

理資料進行病理 AI 模型的開發與訓練，本研究將進一步以臺北榮民總醫院的資料進行獨立驗證，並確保 AI 模型的泛化能力與臨床應用價值。透過病理 AI 模型量化數據與顯示判讀結果，達到輔助病理醫師診斷，並減少醫師彼此間的判讀差異，以提升診斷一致性與準確性。

本研究的預期成果包括：

1. 模型偵測準確度：評估 AI 模型自動偵測肝臟病理特徵的準確性。
2. 數據與病理報告的一致性：分析模型數據與病理診斷的一致性。
3. 診斷輔助效果：驗證 AI 輔助診斷對提高醫師診斷準確度與一致性的成效，並評估其臨床應用潛力。

最終目標為提升診斷效率與精準度，促進個體化治療，進而改善患者的預後及生活品質。

四、試驗方法

(一) 受試者標準

- 納入標準：在本院接受肝臟組織檢查的患者，檢體類型包括穿刺及手術檢體，病理染色類型涵蓋 H&E 染色與 Masson 染色。
- 排除標準：
 - 年齡低於18歲。
 - 病理玻片損傷，導致影像無法清晰呈現。
- 受試者樣本數：
 - 脂肪肝：H&E 及 Masson 染色各200例。
 - 肝腫瘤：H&E 染色200例。
- 招募方式：利用回溯性資料，不需新病例招募。

(二) 試驗設計及進行方法

1. 模型偵測準確度驗證使用
 - AI 偵測切割影像中的關鍵組織，並比對醫師修正結果以計算準確度。
 - 指標：依模型選擇 F1-score、mIoU 等評分指標。
2. 模型數據與病理報告的相關性分析
 - 分析模型生成數據與疾病分級之相關性，計算 AI 對疾病等級的預測準確性。
3. 模型輔助診斷效果驗證
 - 三階段驗證：第一階段醫師獨立診斷，第二階段參考 AI 結果，第三階段醫師共同討論後達成共識。
 - 指標：比較醫師診斷的準確性與一致性。

(三) 評估及統計方法

- 評估方式：準確度、F1-score、mIoU 等指標。
- 相關性分析：使用皮爾森相關係數計算模型數據與病理報告的關聯性。
- 診斷一致性評估：以 Cohen's Kappa 系數計算醫師間診斷一致性。

五、 資料隱私保護

本研究計畫收集資料包含數位病理影像及病理報告，報告資訊涵蓋年齡、性別、CRN 分數腫瘤期別、治療方式、存活資料及檢體等。資料將儲存於臺北榮民總醫院病理檢驗部，成大電機系詹寶珠老師實驗室則取得副本進行研究，副本資料儲存於研究室獨立建置之 NAS 系統，該 NAS 不與外部網路連接，並僅允許經授權的計畫主持人、研究人員進行存取。此外，所有病人檢體皆重新編碼為代號，資料中不包含姓名、病歷號碼等任何個人識別資訊；研究計畫結束後，去識別化的資料會永久保存於國立成功大學詹寶珠教授實驗室作於學術研究使用，不對外公開。

本研究雖然不涉及與受試者的直接接觸，但所使用的病理影像和病灶報告中可能包含病患的個人資訊。為保護受試者的隱私，本研究可提供相關的去識別化程式，讓所有資料在提供給研究人員前，就已進行嚴格的去識別化處理，以確保任何可識別的個人資訊均不會出現在研究結果中。具體過程如下：

1. 檢體取得與去識別化處理：病理部醫師從資料庫中提取檢體，並對玻片與病灶報告進行去識別化處理。此過程包括重新編碼及移除所有可識別的個人資訊（如姓名、病歷號碼等）。完成去識別化後，處理後的去識別化資料將交予研究人員進行後續分析。
2. 資料處理與儲存：所有病理影像及病灶報告在去識別化後，不再包含病患姓名、病歷號碼或其他個人識別資訊，並將副本存儲於國立成功大學電機工程學系詹寶珠教授實驗室的 NAS 系統中。該系統與外部網路隔離，僅限授權人員存取，以確保資料安全。

透過上述流程，確保所有病理影像及病灶報告在資料處理與分析過程中均符合嚴格的隱私保護要求。

六、 預期試驗效果

本研究預期達成以下成果：

- 提升病理診斷準確性：透過 AI 技術提高影像判讀準確性，提供一致的診斷結果。
- 增強診斷一致性與客觀性：透過量化數據，減少診斷主觀性，確保醫師間診斷一致性。

- 縮短診斷時間，提升效率：AI 自動分析加快診斷過程，減少醫師工作負擔。
- 驗證 AI 的臨床應用潛力：多階段驗證模型於臨床應用的成效。
- 提供量化數據輔助精準醫療：AI 數據可用於疾病預後和治療效果評估，支持精準醫療。

最終，本研究期望證明 AI 技術對肝臟病理診斷的應用價值，提升診斷精準度，推動精準醫療，改善患者治療效果及生活品質。

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